Chapter 1

Reverse-time diffusion and generation

A diffusion model generates by running the channel of **??** backwards in time, and the only ingredient the backward equation needs that the forward one does not is the score. This chapter derives the reverse-time equation (Anderson's theorem) from the Fokker–Planck equation, explains why an error in the score is exactly what an error in the generated law is made of, writes down the Euler–Maruyama scheme the code uses, and then describes the paper's generation experiment E05: its same-Brownian-path design, its arms, and every metric by which the samples are judged, including the transition-law metric (Hellinger distance) used in the learning section.

1.1 Anderson's reverse-time equation

Consider a forward diffusion on $[0, T]$,

$$\mathrm{d}\mathbf{x}_t = f(\mathbf{x}_t, t)\ \mathrm{d}t + g(t)\ \mathrm{d}\mathbf{W}_t, \qquad \mathbf{x}_0 \sim P_0, \tag{1.1}$$

with smooth positive marginal densities p_t. The evolution of p_t is the **Fokker–Planck equation**

$$\partial_t p_t = -\nabla \cdot (f\, p_t) + \tfrac{1}{2} g^2\, \Delta p_t, \tag{1.2}$$

where $\nabla\cdot$ is the divergence and $\Delta = \nabla \cdot \nabla$ the Laplacian (not to be confused with Δ_t). The reader who has not met Equation (1.2) may read it as a conservation law: the drift transports probability with velocity f, and the noise spreads it by diffusion with coefficient $\frac{1}{2}g^2$.

Theorem 1.1 (Time reversal)**.** *Let $\tau = T - t$ and let $\bar{\mathbf{x}}_\tau$ solve*

$$\mathrm{d}\bar{\mathbf{x}}_\tau = \left[-f(\bar{\mathbf{x}}_\tau, T-\tau) + g(T-\tau)^2\, \nabla \log p_{T-\tau}(\bar{\mathbf{x}}_\tau)\right] \mathrm{d}\tau + g(T-\tau)\ \mathrm{d}\bar{\mathbf{W}}_\tau, \qquad \bar{\mathbf{x}}_0 \sim p_T, \tag{1.3}$$

with $\bar{\mathbf{W}}$ a Brownian motion. Then $\bar{\mathbf{x}}_\tau$ has the law $p_{T-\tau}$ for every $\tau \in [0, T]$; in particular $\bar{\mathbf{x}}_T \sim P_0$. Anderson (1982) proved more: the process $t \mapsto \mathbf{x}_{T-t}$ itself is a solution of Equation (1.3)*.*

Proof of the marginal statement. Scalar case, and write $p = p_t(x)$. Since $\frac{1}{2}g^2\partial_x^2 p = \partial_x(g^2\partial_x p) - \frac{1}{2}g^2\partial_x^2 p$ and $\partial_x p = p\,\partial_x \log p$, the Fokker–Planck equation can be rewritten as

$$\partial_t p = -\partial_x\Big[(f - g^2\partial_x \log p)p\Big] - \tfrac{1}{2} g^2\, \partial_x^2 p.$$

Now set $q(x, \tau) = p_{T-\tau}(x)$, so that $\partial_\tau q = -\partial_t p$:

$$\partial_\tau q = -\partial_x\Big[(-f + g^2\partial_x \log q)q\Big] + \tfrac{1}{2} g^2\, \partial_x^2 q.$$

This is the Fokker–Planck equation of the diffusion with drift $-f + g^2\partial_x \log q$ and diffusion coefficient g, run in the time τ, and $q(\cdot, 0) = p_T$. The solution of Equation (1.3) has marginals obeying the same equation with the same initial condition, hence the same marginals, by uniqueness for this parabolic equation. In L dimensions replace ∂_x by $\nabla\cdot$ and ∂_x^2 by Δ line by line. □

A useful by-product of the same rewriting is the **probability-flow ODE**: $\frac{1}{2}g^2\Delta p = \nabla\cdot(\frac{1}{2}g^2 p\nabla\log p)$ turns Equation (1.2) into $\partial_t p = -\nabla\cdot[(f-\frac{1}{2}g^2\nabla\log p)p]$, the transport equation of the deterministic flow $\dot{\mathbf{x}} = f - \frac{1}{2}g^2\nabla\log p_t$, which has the same marginals as the SDE and is implemented in `probability_flow_ode` of `src/markov_score/reverse.py`.

Corollary 1.2 (The channel reversed)**.** *For the channel $f(\mathbf{x},t) = -\mathbf{x}$, $g = \sqrt{2}$, the reverse equation in the original time t running from T down to 0 is*

$$\mathrm{d}\mathbf{x} = \big[-\mathbf{x} - 2\,S(\mathbf{x},t)\big]\,\mathrm{d}t + \sqrt{2}\;\mathrm{d}\bar{\mathbf{W}}_t \qquad (\,\mathrm{d}t < 0), \tag{1.4}$$

that is, in the forward-running time τ, $\mathrm{d}\bar{\mathbf{x}} = [\bar{\mathbf{x}} + 2S(\bar{\mathbf{x}}, T-\tau)]\,\mathrm{d}\tau + \sqrt{2}\;\mathrm{d}\bar{\mathbf{W}}_\tau$.

The drift has two parts. The term $\bar{\mathbf{x}}$ undoes the contraction of the forward drift; the term $2S$ pushes towards regions where P_t is large. At high noise, where $S \approx -\mathbf{x}$, the two nearly cancel and the reverse process is again an Ornstein–Uhlenbeck process (Example 1.7); at low noise the score carries all the structure of P_0.

1.2 Why the score is exactly what drives generation

Everything in Equation (1.4) is known except S: the drift $-\mathbf{x}$ and the coefficient $\sqrt{2}$ are the channel's. A generator that replaces S by an estimate $\widehat{s}$ runs a different diffusion and produces a different law, and the discrepancy is controlled by the score error alone.

Proposition 1.3 (Score error controls the generated law)**.** *Let $\mathbb{P}$ be the law of the paths of Equation (1.4) on $[t_{\min}, T]$ started at $\mathbf{x}_T \sim P_T$, and $\mathbb{Q}$ the law of the paths of the same equation with $\widehat{s}$ in place of S and started at $\gamma = \mathcal{N}(0,\mathbf{I})$. Then*

$$D_{\mathrm{KL}}\big(P_{t_{\min}} \,\|\, \mathbb{Q}_{t_{\min}}\big) \le D_{\mathrm{KL}}\big(P_T \,\|\, \gamma\big) + \int_{t_{\min}}^{T} \mathbb{E}_{P_t}\big|\widehat{s}(\mathbf{x}_t,t) - S(\mathbf{x}_t,t)\big|^2\,\mathrm{d}t, \tag{1.5}$$

where $\mathbb{Q}_{t_{\min}}$ is the law of the generated sample at time $t_{\min}$.

Proof sketch. Two diffusions with the same diffusion coefficient g and drifts μ_1, μ_2 have path laws whose Kullback–Leibler divergence is, by Girsanov's theorem, $\frac{1}{2}\int \mathbb{E}|\mu_1 - \mu_2|^2/g^2\;\mathrm{d}t$ plus the divergence of the initial laws. Here $\mu_1 - \mu_2 = 2(S - \widehat{s})$ and $g^2 = 2$, so the integrand is $\frac{1}{2}\cdot 4|S-\widehat{s}|^2/2 = |S-\widehat{s}|^2$, with the expectation under the true path law, whose marginal at time t is P_t. The divergence of the marginals at $t_{\min}$ is at most that of the path laws (data processing). □

This is Theorem 1 of Song, Durkan, Murray and Ermon (2021). It says that the time-integrated mean squared score error under the true noised laws is the quantity a generator pays for, in nats. The paper's estimand (**??**) is the square root of that integrand summed over the twelve levels and divided by the reference energy; the generation experiment E05 measures whether the ordering of estimators by score error is the ordering by sample quality, which Equation (1.5) predicts but does not guarantee, because it is an upper bound.

1.3 The Euler–Maruyama scheme and its grid

Definition 1.4 (Euler–Maruyama for the reverse channel)**.** Given a decreasing sequence of times $T = t_0 > t_1 > \cdots > t_n = t_{\min}$ with steps $h_k = t_k - t_{k+1} > 0$, the scheme is

$$\mathbf{x}_{k+1} = \mathbf{x}_k + h_k\big[\mathbf{x}_k + 2\,\widehat{s}(\mathbf{x}_k, t_k)\big] + \sqrt{2h_k}\,\xi_k, \qquad \xi_k \sim \mathcal{N}(0,\mathbf{I})\ \text{i.i.d.}, \qquad \mathbf{x}_0 \sim \mathcal{N}(0,\mathbf{I})\,. \tag{1.6}$$

This is `reverse_sde` and `reverse_sde_with_increments` in `src/markov_score/reverse.py`: the drift is evaluated at the left endpoint, and the Brownian increment over the step is $\sqrt{h_k}\,\xi_k$, scaled by the $\sqrt{2}$ of the channel. Three choices need explaining.

The grid is geometric. The code uses $t_k = t_{\max}\, r^k$ with $r = (t_{\min}/t_{\max})^{1/n}$ (`time_grid`), so that $h_k = (1-r)\, t_k$: the step is a fixed fraction of the current time. The reason is that the reverse drift can be stiff at small t: by **??**, $\nabla S = -\mathbf{I}/\Delta_t + (e^{-2t}/\Delta_t^2)\,\mathrm{Cov}(\mathbf{a} \mid \mathbf{x})$, whose entries can be of size $1/\Delta_t \approx 1/(2t)$ when the posterior is narrower than the likelihood (a kinked or discrete prior), and an explicit scheme is only stable for steps below the reciprocal of the stiffness. A step proportional to t keeps $h_k/\Delta_{t_k} \approx (1-r)/2$ fixed. With the paper's $t_{\max} = 3$, $t_{\min} = 0.02$ and $n = 400$ steps (`experiments/e05_generation.py`), $\ln(t_{\min}/t_{\max}) = -5.0106$, $r = e^{-0.012527} = 0.98755$, the first step is $h_0 = 3 \times 0.01245 = 0.0374$ and the last is $h_{399} = 0.0203 \times 0.01245 = 0.00025$; the ratio h_k/Δ_{t_k} is about 0.006 throughout.

The start is $\mathcal{N}(0, \mathbf{I})$, not P_T. The true law at $T = 3$ is not available to sample from, and γ is its limit (**??**). The cost is the first term of Equation (1.5). For the covariance twin it is computable: $\mathbf{\Sigma}_T = \mathbf{I} + e^{-2T}(\mathbf{\Sigma}_0 - \mathbf{I})$ and, to second order in e^{-2T}, $D_{\mathrm{KL}}(\mathcal{N}(0, \mathbf{\Sigma}_T) \,\|\gamma) = \frac{1}{2}[\mathrm{tr}\, \mathbf{\Sigma}_T - L - \log\det \mathbf{\Sigma}_T] \approx \frac{1}{4} e^{-4T} \|\mathbf{\Sigma}_0 - \mathbf{I}\|_F^2$. For the frozen chain, $\|\mathbf{\Sigma}_0 - \mathbf{I}\|_F^2 = 2\sum_{k=1}^{31}(32-k)\rho^{2k} \approx 148$ at $\rho = 0.85$ (a geometric sum), so the start costs about $\frac{1}{4} e^{-12} \cdot 148 \approx 2 \times 10^{-4}$ nats: negligible next to the discretisation, and the reason $t_{\max} = 3$ suffices.

The stop is $t_{\min} = 0.02$, not 0. At $t = 0$ the score of a law with a density that is not smooth (the Laplace chain has a kink) is singular, and for a law without a density it does not exist. The generated sample is therefore a sample at $t_{\min}$ and is compared with a reference sample of the true law at $t_{\min}$, obtained by drawing clean sequences and noising them once (`rng_ref` in `_gen_cell`); comparing with clean sequences would be a category error. The alternative readout $\langle \mathbf{a} \rangle_{x,t} = e^t(\mathbf{x} + \Delta_t S)$ of **??** (`denoising_readout`) returns a posterior mean, which is not a sample.

Order of accuracy. For an additive noise, as here, Euler–Maruyama has strong order one (the pathwise error is $O(h)$) and weak order one (errors of expectations are $O(h)$); the discretisation bias is measured, not assumed, by the step study below.

1.4 The same-Brownian-path design of E05

The experiment (`experiments/e05_generation.py`) is organised in cells (seed, M). In each cell one initial sample $\mathbf{x}_0$ of 2000 sequences and one array of Brownian increments for all 400 steps are drawn once, and every arm is integrated by `reverse_sde_with_increments` on exactly those increments. The arms differ only in the score function $\widehat{s}$:

- *exact*: the tree-exact score under the true kernel (grid message passing; its distance to the true law is the sampler's own discretisation floor);
- *em–bp*: the mixture kernel fitted in E02a for the same seed and M, at its validation-selected checkpoint, driven through the same message passing;
- *lmmse*: the covariance twin, $\widehat{s} = -\mathbf{\Sigma}_t^{-1}\mathbf{x}$ (**??**);
- *isotropic*: $\widehat{s} = -\mathbf{x}$, the score of γ;
- *net*: network checkpoints from E03's confirmatory part for the same cell, converted from posterior means by **??**.

Because the noise is shared, the difference between two arms' samples is due to their scores and not to Monte Carlo scatter, and the metrics can be compared within cell and then paired across seeds, as everywhere else in the paper. The reference sample R of 4000 sequences at $t_{\min}$ is drawn from its own stream. The *step* part of the experiment integrates the exact arm at 100, 200, 400 and 800 steps on **nested Brownian paths**: the increments of the finest grid are drawn once and the coarse grid's increment over an interval is the sum of the fine increments inside

it (`nested_brownian_path`, `coarsen_increments`). Two resolutions are then driven by the same Brownian path and the root mean square of their difference is a **strong error**, a pure measure of discretisation. With independent noise at each resolution the difference would be dominated by sampling scatter and the study would show only robustness, which is the mistake the code's docstring records.

1.5 The sample-quality metrics

Let $G \in \mathbb{R}^{n_g \times L}$ be the generated sample and $R \in \mathbb{R}^{n_r \times L}$ the reference sample, both at $t_{\min}$. All definitions are those of `src/markov_score/gen_metrics.py` (`compare`); the paper's generation table prints the first, the second, the fourth and the sliced Wasserstein distance.

Residuals. For a sample Y the **autoregressive residuals** are $r_{j,i} = y_{j,i+1} - \rho\, y_{j,i}$ with the true ρ, $i = 1, \dots, L-1$. At $t_{\min} = 0$ they would be the innovations ε_{i+1}; at $t_{\min} > 0$ they are $e^{-t_{\min}}\varepsilon_{i+1} + \sqrt{\Delta_{t_{\min}}}(z_{i+1} - \rho z_i)$.

Excess kurtosis. For a pooled sample $\{u\}$ with central moments m_2, m_4, $\widehat{\kappa} = m_4/m_2^2 - 3$. Applied to the residuals it measures the non-Gaussian shape of the innovation that the sampler reproduces; a Gaussian-innovation sampler (the twin) gives zero up to sampling error. Its standard error is a bootstrap over sequences (200 resamples).

Proposition 1.5 (Residual kurtosis at $t_{\min}$)**.** *If the innovation has variance q and excess kurtosis κ, then the residual of the true law at level t has*

$$\mathrm{Var}(r) = e^{-2t}q + \Delta_t(1+\rho^2), \qquad \mathrm{cum}_4(r) = e^{-4t}\kappa\, q^2, \qquad \textit{excess kurtosis} = \frac{\kappa\, e^{-4t} q^2}{\left(e^{-2t}q + \Delta_t(1+\rho^2)\right)^2}.$$

Proof. $r = e^{-t}\varepsilon + \sqrt{\Delta_t}(z' - \rho z)$ is a sum of independent terms; variances add, and fourth cumulants add with the Gaussian terms contributing nothing. □

So the reference row of the paper's table does not show the innovation's excess kurtosis 3 but its attenuated and diluted value at $t_{\min}$; the worked example computes it. The same holds for every arm, which is why the table is read as a comparison with the reference row and not with 3.

Lag fourth cumulant. $\mathrm{cum}(r_i, r_i, r_{i+1}, r_{i+1}) = \mathbb{E}[r_i^2 r_{i+1}^2] - \mathbb{E}[r_i^2]\mathbb{E}[r_{i+1}^2] - 2\mathbb{E}[r_i r_{i+1}]^2$ for centred residuals, pooled over sites. For the true law it is exactly zero at every t: r_i and r_{i+1} share only the Gaussian z_{i+1}, joint cumulants are additive over independent components, and a Gaussian has no fourth cumulant. It is therefore a check for spurious dependence between neighbouring residuals, which a sampler with a wrong score can create; it is not a signature of the innovation's shape.

Marginal Wasserstein-1. Pooling all entries of G and of R into two one-dimensional samples with quantile functions F_G^{-1}, F_R^{-1}, $W_1 = \int_0^1 |F_G^{-1}(u) - F_R^{-1}(u)|\,\mathrm{d}u$, approximated by the mean over 1999 equally spaced u. It compares the site marginals only.

Sliced Wasserstein-2. For 256 random unit vectors θ_k (common seed), project both samples, $G\theta_k$ and $R\theta_k$, compute the one-dimensional W_2^2 of the projections by quantile matching, $W_2^2(\mu,\nu) = \int_0^1 (F_\mu^{-1} - F_\nu^{-1})^2\,\mathrm{d}u$, and report $\left(\frac{1}{256}\sum_k W_2^2\right)^{1/2}$. It sees the joint law through its one-dimensional projections, and is dominated by second-order structure: two laws with the same covariance and different fourth cumulants project to nearly the same one-dimensional laws, which is why the twin and the exact arm are close in this column.

Energy distance. $2\mathbb{E}|G-R| - \mathbb{E}|G-G'| - \mathbb{E}|R-R'|$ with independent copies, estimated on subsamples of 1000 (Székely and Rizzo); it is zero if and only if the laws coincide, and is again dominated by low-order structure in practice.

Covariance errors. The sample covariance $\widehat{\boldsymbol{\Sigma}}$ of G against a target $\boldsymbol{\Sigma}$: the relative Frobenius error $\|\widehat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}\|_F / \|\boldsymbol{\Sigma}\|_F$ and the worst absolute error of the mean diagonal at lags 0 to 5. The code passes $\boldsymbol{\Sigma}_0$ as the target although the sample lives at $t_{\min}$, where the covariance is $\boldsymbol{\Sigma}_{t_{\min}} = e^{-2t_{\min}}\boldsymbol{\Sigma}_0 + \Delta_{t_{\min}}\mathbf{I}$; the difference is $\Delta_{t_{\min}}(\boldsymbol{\Sigma}_0 - \mathbf{I})$, whose relative Frobenius size at $t_{\min} = 0.02$ is about $0.039 \times \|\boldsymbol{\Sigma}_0 - \mathbf{I}\|_F / \|\boldsymbol{\Sigma}_0\|_F \approx 0.035$ for the frozen chain, a floor the exact arm cannot go below. The paper's table does not print this column.

Residual autocorrelation. $\rho_k = \mathbb{E}[r_i r_{i+k}]/\mathbb{E}[r_i^2]$ for $k = 1, \dots, 5$ on the pooled centred residuals; the reported number is the largest absolute difference between the generated and the reference values. For the true law at $t_{\min}$ the residuals are not white ($\rho_1 = -\rho\Delta_{t_{\min}}/\operatorname{Var}(r)$, from the shared z_{i+1}), which is why the reference values are subtracted rather than zero.

Histogram divergence of the residuals. With 60 bins between the 0.001 and 0.999 quantiles of the reference residuals, p the reference histogram and q the generated one with add-one smoothing, $\sum_{p_b>0} p_b \log(p_b/q_b)$: a Kullback–Leibler divergence of the innovation shape, true against generated.

Log-likelihood under the true law. For every sequence of G and of R the evidence $\log P_{t_{\min}}(\mathbf{x})$ under the true kernel is computed exactly for the discretised model by the same message passing that gives the score (the accumulated normalisation constants of the recursion; the function `log_likelihood_true`), and the reported gap is

$$\Delta\ell = \frac{1}{n_r}\sum_{R} \log P_{t_{\min}}(\mathbf{x}) - \frac{1}{n_g}\sum_{G} \log P_{t_{\min}}(\mathbf{x}). \tag{1.7}$$

Writing P for the true law and Q for the generated one, in expectation $\Delta\ell = \mathbb{E}_P \log P - \mathbb{E}_Q \log P = D_{\mathrm{KL}}(Q\|P) + H(Q) - H(P)$ with H the differential entropy: a difference of cross-entropies, not a divergence. It is zero when $Q = P$, positive when Q is misplaced or too spread, and can be negative when Q is more concentrated than P on the high-density region. The exact arm's value is the discretisation floor of the sampler, since its only error is the scheme's; the superseded four-seed run recorded a negative value for that arm, which is consistent with the definition and is the reason the sign is worth stating.

1.6 A transition-law metric: the Hellinger distance

The learning section of the paper reports that the transition law is recovered "to Hellinger distance" a stated value at the lowest level. The metric is the function `transition_hellinger` of `src/markov_score/metrics.py`.

Definition 1.6 (Hellinger distance)**.** For two densities p, q on $\mathbb{R}$,

$$H^2(p,q) = \tfrac{1}{2}\int \left(\sqrt{p} - \sqrt{q}\right)^2 \mathrm{d}u = 1 - \int \sqrt{pq}\ \mathrm{d}u \in [0,1],$$

the second form because $\int p = \int q = 1$. On the quadrature grid with weights w_k the integral is $\frac{1}{2}\sum_k w_k(\sqrt{p_k} - \sqrt{q_k})^2$ after each column has been normalised under the same weights.

For a transition kernel every parent state u_j gives one density $K(\cdot \mid u_j)$, hence one distance H_j between the fitted and the true kernel. The code reports the median, mean, 90th percentile and maximum over parents, the same over the interior parents $|u_j| \le A/2$ (near the edge of the domain the true column has mass outside the grid and the distance picks up truncation), and versions weighted by the parent law, which defaults to the standard normal because $\mathrm{Var}(a_i) = 1$: the weighted mean $\sum_j \pi_j H_j$, the weighted median, and $\sum_j \pi_j H_j^2$. The second form of the definition is the one the code evaluates, because $1 - \int \sqrt{pq}$ subtracts two numbers near one and loses everything below 10^{-15} to cancellation; the first form returns exactly zero for identical columns. Which of these summaries the paper's sentence quotes is not said in the paper; the reader should take the weighted mean as the intended one and check the generator.

1.7 Worked examples

Example 1.7 (The reverse of a Gaussian, and the bias of a step)**.** Let $P_0 = \mathcal{N}(0,1)$ in one dimension, so that $S(x,t) = -x$ at every t. The reverse equation Equation (1.4) in the time τ is $\mathrm{d}\bar{x} = [\bar{x} - 2\bar{x}]\,\mathrm{d}\tau + \sqrt{2}\,\mathrm{d}\bar{W} = -\bar{x}\,\mathrm{d}\tau + \sqrt{2}\,\mathrm{d}\bar{W}$: the forward channel again, for which $\mathcal{N}(0,1)$ is stationary, as it must be. The Euler step Equation (1.6) with constant h is $x \leftarrow (1-h)x + \sqrt{2h}\,\xi$, and the variance v of the iterate obeys $v \leftarrow (1-h)^2 v + 2h$, whose fixed point is

$$v^* = \frac{2h}{1-(1-h)^2} = \frac{2h}{2h-h^2} = \frac{2}{2-h} = 1 + \tfrac{h}{2} + O(h^2).$$

The scheme's stationary variance is biased by $h/2$: with the first geometric step $h_0 = 0.0374$ the bias would be 1.9%, with the last, 0.01%. This is the weak error of order one, and it is what the exact arm's log-likelihood gap measures on the chain.

Example 1.8 (The residual kurtosis of the frozen chain at $t_{\min}$)**.** Take $\rho = 0.85$, $q = 1 - \rho^2 = 0.2775$, a Laplace innovation ($\kappa = 3$) and $t_{\min} = 0.02$, so $e^{-2t_{\min}} = 0.96079$, $e^{-4t_{\min}} = 0.92312$, $\Delta_{t_{\min}} = 0.03921$. By Proposition 1.5,

$$\mathrm{Var}(r) = 0.96079 \times 0.2775 + 0.03921 \times 1.7225 = 0.26662 + 0.06754 = 0.33416,$$

$$\mathrm{cum}_4(r) = 3 \times 0.92312 \times 0.2775^2 = 3 \times 0.92312 \times 0.07701 = 0.21327,$$

and the excess kurtosis is $0.21327/0.33416^2 = 0.21327/0.11166 = 1.91$. A perfect sampler at $t_{\min} = 0.02$ should therefore show a residual excess kurtosis near 1.9, not 3; the paper's reference row is to be read against this value (see the paper, Section 5, the generation table). The Gaussian twin shows 0; a sampler that reproduces only part of the fourth cumulant lands in between.

Example 1.9 (Hellinger distances one can compute by hand)**.** For $p = \mathcal{N}(0,1)$ and $q = \mathcal{N}(0,\sigma^2)$, $\int \sqrt{pq} = \sqrt{2\sigma/(1+\sigma^2)}$, so at $\sigma = 1.1$: $2.2/2.21 = 0.99548$, square root 0.99774, $H^2 = 0.00226$, $H = 0.048$. For $p = \mathcal{N}(0,1)$ and $q = \mathcal{N}(\mu,1)$, $\int \sqrt{pq} = e^{-\mu^2/8}$, so at $\mu = 0.5$: $H^2 = 1 - e^{-0.03125} = 0.0308$, $H = 0.175$. A ten per cent error in the width of an innovation costs less than a half-unit shift of its centre; a column-wise Hellinger summary is sensitive to both.

Where it is used. Generation is Section 5 of the paper ("Generation") with its table, and the reverse SDE is named there with its step count, grid and shared Brownian path; the Hellinger sentence is in Section 4. Experiment E05 (parts `generate` and `steps`); the arms come from E02a (fitted kernels) and E03 (network checkpoints). Code: `src/markov_score/reverse.py` (`time_grid`, `reverse_sde`, `reverse_sde_with_increments`, `nested_brownian_path`, `coarsen_increments`, `probability_flow_ode`, `denoising_readout`), `src/markov_score/gen_metrics.py` (`compare` and the functions it calls), `src/markov_score/metrics.py` (`transition_hellinger`), `experiments/e05_generation.py`.

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